Lecture 2

Structure of groups: homomorphisms, quotients, and products

We learn two things: to compare groups by structure-preserving maps, and to take them apart into simpler pieces. To that end we meet homomorphisms and their kernels, conjugacy and its geometric meaning, quotient groups and the first isomorphism theorem. The reading carries two further developments: how groups are "assembled" again, by direct and semidirect products, and how a finite group is decomposed into simple constituents.

By the end of this lecture you will be able to:

- Find the kernel and the image of a homomorphism.
- Compute conjugacy classes.
- Form quotient groups and apply the first isomorphism theorem.

2.1 Homomorphisms

Lecture 1 declared two groups "the same" when a bijection matches their multiplication tables. Far more common, and no less useful, are maps that preserve multiplication *without* being bijective. They compress a group onto a smaller one, retaining some structure and forgetting the rest.

Definition 2.1 (Homomorphism)**.** A map $\varphi\colon G \to H$ between groups is a *homomorphism* if

$$\varphi(g_1 g_2) = \varphi(g_1)\,\varphi(g_2) \qquad \text{for all } g_1, g_2 \in G.$$

An *isomorphism* is precisely a bijective homomorphism. The *kernel* and *image* of φ are

$$\ker\varphi = \{\, g \in G : \varphi(g) = e_H \,\}, \qquad \operatorname{im}\varphi = \{\, \varphi(g) : g \in G \,\}.$$

Example 2.2.

(i) The sign map $\operatorname{sgn}\colon S_n \to \{\pm 1\}$ defined in Lecture 1, which sorts permutations into even and odd, is a homomorphism. Its kernel is the subgroup of all even permutations – the alternating group A_n inside S_n.

(ii) The determinant of the group of all invertible $n \times n$ matrices, $\det\colon \mathrm{GL}(n,\mathbb{F}) \to \mathbb{F}\setminus\{0\}$, is a homomorphism, since $\det(AB) = \det A \det B$; its kernel is the *special linear group* $\mathrm{SL}(n,\mathbb{F})$ of matrices of determinant 1.

(iii) Realise D_n as orthogonal transformations of the plane, as in Lecture 1. The restriction

of the determinant,

$$\nu\colon D_n \to \{\pm 1\}, \qquad \nu(g) = \det g = \begin{cases} +1, & g \text{ a rotation}, \\ -1, & g \text{ a reflection}, \end{cases}$$

records the *orientation behaviour* of each symmetry operation. This *orientation homomorphism* will be our running example; its kernel is the rotation subgroup $\langle r\rangle \cong C_n$ (Figure 2.1).

(iv) Reduction modulo n, the map $\mathbb{Z} \to \mathbb{Z}_n$ sending an integer to its residue, is a homomorphism of additive groups with kernel $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$.

(v) The map $\varphi(x) = e^{2\pi i x}$ is a homomorphism from $(\mathbb{R}, +)$ onto the circle group $\mathrm{U}(1) = \{z \in \mathbb{C} : |z| = 1\}$, since $e^{2\pi \mathrm{i}(x+y)} = e^{2\pi \mathrm{i} x} e^{2\pi \mathrm{i} y}$; its kernel is $\mathbb{Z}$.

(vi) For any pair G, H the *trivial* homomorphism sends all of G to e_H: kernel G, image $\{e_H\}$. ◁

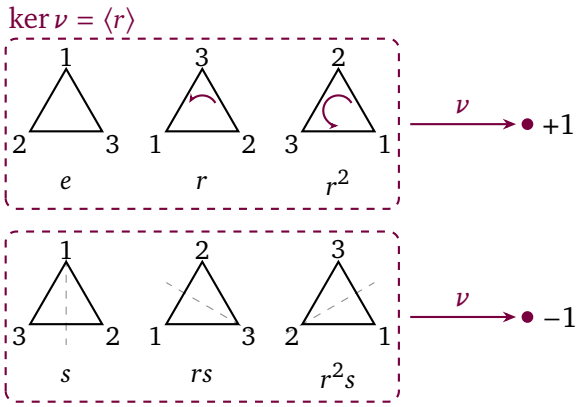


Figure 2.1: The orientation homomorphism $\nu\colon D_3 \to \{\pm 1\}$. All three rotations (top) go to $+1$ and all three reflections (bottom) to -1: each dashed block is collapsed to a single point, and the top block is the kernel. Under the isomorphism $D_3 \cong S_3$ this is the sign sgn: the rotations are the even permutations, the reflections the odd ones.

Proposition 2.3 (Basic properties). Let $\varphi\colon G \to H$ be a homomorphism. Then:

(i) $\varphi(e_G) = e_H$, and $\varphi(g^{-1}) = \varphi(g)^{-1}$; more generally $\varphi(g^k) = \varphi(g)^k$ for all $k \in \mathbb{Z}$. If g has finite order, then $\operatorname{ord}(\varphi(g))$ divides $\operatorname{ord}(g)$;

(ii) $\operatorname{im}\varphi$ is a subgroup of H, and $\ker\varphi$ is a subgroup of G;

(iii) φ is injective if and only if $\ker\varphi = \{e_G\}$;

(iv) composites of homomorphisms are homomorphisms.

Proof. (i) Since $\varphi(e)\varphi(e) = \varphi(ee) = \varphi(e)$, cancellation in H gives $\varphi(e) = e_H$. Then $\varphi(g)\varphi(g^{-1}) = \varphi(gg^{-1}) = e_H$ identifies $\varphi(g^{-1})$ as the inverse of $\varphi(g)$. The identity $\varphi(g^k) = \varphi(g)^k$ follows by induction in both directions (for positive k from $\varphi(g^{k+1}) = \varphi(g^k)\,\varphi(g)$, and for negative from $\varphi(g^{-k}) = \varphi\big((g^k)^{-1}\big) = \varphi(g^k)^{-1}$). If g has finite order and $n = \operatorname{ord}(g)$ then $\varphi(g)^n = \varphi(g^n) = e_H$, so $\operatorname{ord}(\varphi(g)) \mid n$ by the order lemma of Lecture 1 (whose "if and only if" used only that the element has finite order, not that the group is finite).

(ii) $\operatorname{im}\varphi$ contains e_H and is closed under products and (by (i)) inverses; likewise if $\varphi(g) = \varphi(k) = e_H$ then $\varphi(gk) = e_H$ and $\varphi(g^{-1}) = e_H$. The subgroup test of Lecture 1 applies.

(iii) If φ is injective, only e_G can map to e_H. Conversely suppose $\ker\varphi = \{e\}$ and $\varphi(g) = \varphi(g')$; then $\varphi(g^{-1}g') = \varphi(g)^{-1}\varphi(g') = e_H$, so $g^{-1}g' = e$, i.e. $g = g'$.

(iv)
$$\begin{aligned}(\psi\varphi)(g_1g_2) &= \psi\big(\varphi(g_1g_2)\big) = \psi\big(\varphi(g_1)\varphi(g_2)\big)\\ &= \psi(\varphi(g_1))\,\psi(\varphi(g_2)) = (\psi\varphi)(g_1)\,(\psi\varphi)(g_2).\end{aligned}$$

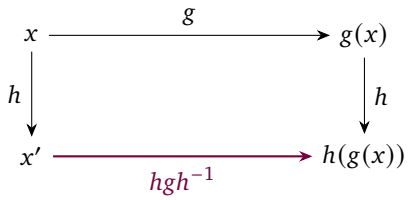
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A homomorphism thus forgets information in a controlled way, and the kernel measures exactly *how much* is forgotten: trivial kernel, nothing forgotten; kernel $\langle r\rangle$ in the orientation map, everything forgotten except whether the operation flips the plane over. Making "G with $\ker\varphi$ forgotten" into a group in its own right is the business of Section 2.3. But first we need the right notion of how a group acts on itself. For that we define the group-theoretic analogue of a "similarity transformation".

2.2 Conjugacy

Definition 2.4 (Conjugacy)**.** Elements g, g' of a group G are *conjugate* if $g' = hgh^{-1}$ for some $h \in G$. Conjugacy is an equivalence relation ($g = ege^{-1}$; if $g' = hgh^{-1}$ then $g = h^{-1}g'h$; and $k(hgh^{-1})k^{-1} = (kh)g(kh)^{-1}$), so G is partitioned into *conjugacy classes*; the class of g is denoted $[g] = \{hgh^{-1} : h \in G\}$.

Conjugation has a sharp physical interpretation: hgh^{-1} is *the operation g as seen in the frame transformed by h*. Suppose G consists of transformations of space, and write $x' = h(x)$. Then the identity
$$(hgh^{-1})(x') = h(g(x))$$
holds. It says that hgh^{-1} does to the transformed point x' exactly what g does to x: first transform back with h^{-1}, then apply g, then transform forward again with h. On the right-hand side the same result $g(x)$ is carried into the transformed frame by h. Both routes from x to $h(g(x))$ agree:



Conjugate operations are therefore *the same kind of operation*, differing only in the frame in which they are written. For instance a rotation through θ about one axis is conjugate to the rotation through θ about any other axis the group can reach by its action. Quantities that do not depend on the frame are therefore constant on conjugacy classes. *Characters*, which we meet in Lecture 5, are of this kind.

Example 2.5 (Conjugation in D_n)**.** From the relations $r^n = s^2 = e$, $sr^ks^{-1} = r^{-k}$ of Lecture 1:

(i) $s\,r^k\,s^{-1} = r^{-k}$: viewed in a mirror, a rotation runs backwards, so r^k and r^{-k} are conjugate;

(ii) $r^m\,(r^j s)\,r^{-m} = r^{j+m}(sr^{-m}) = r^{j+m}\,r^m s = r^{j+2m}s$: conjugating a reflection by the rotation r^m gives the reflection whose axis is turned by $2\pi m/n$ from the original axis, the same angle as r^m itself. The index nevertheless goes up by $2m$, because neighbouring mirror axes are only π/n apart (Figure 2.2);

(iii) $s\,(r^j s)\,s^{-1} = (sr^j)(ss^{-1}) = sr^j = r^{-j}s$, by the same rules: reflections conjugate to reflections in the "mirror" direction. ◁

[self-study]

Example 2.6 (Conjugacy classes of D_5)**.** Start with the rotations. Example 2.5(i) gives $[r] = \{r, r^4\}$ and $[r^2] = \{r^2, r^3\}$: conjugating the rotation r^k by another rotation does nothing (since rotations commute), and conjugating it by the reflection s inverts r^k. Conjugation by r raises the index j of the reflection $r^j s$ by two: $j \mapsto j+2$. Since 5 is odd, shifts by two run through all five residues, so all five reflections form a single class. Hence

$$D_5 \;=\; \{e\} \;\cup\; \{r, r^4\} \;\cup\; \{r^2, r^3\} \;\cup\; \{s, rs, r^2s, r^3s, r^4s\}, \qquad 10 = 1+2+2+5.$$

Geometrically the reflection class is plain: every axis of the regular pentagon passes through one vertex and one edge midpoint — all five axes look alike. ◁

Remark 2.7 (D_4, and even against odd)**.** What decided the matter was that n was odd. For even n shifts by two never turn an even index into an odd one, so the reflections split into *two* classes. Geometry shows the same thing (Figure 2.2): the axes through opposite vertices and the axes through opposite edge midpoints are different kinds of mirror. For the square,

$$D_4 \;=\; \{e\} \cup \{r^2\} \cup \{r, r^3\} \cup \{s, r^2s\} \cup \{rs, r^3s\}\ :$$

five conjugacy classes — a number worth remembering when we count the irreducible representations of D_4 in Lecture 5.

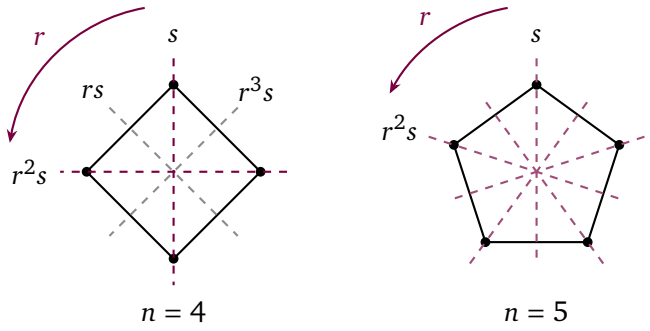


Figure 2.2: Conjugation by the rotation r in D_4 and D_5. A mirror axis turns through the same angle $2\pi/n$ as r, but neighbouring axes are only π/n apart, so the index of the reflection goes up by two: $rsr^{-1} = r^2 s$. In the square an axis through vertices (coloured) therefore always goes to an axis through vertices, and an axis through edge midpoints (grey) to another of its kind: the reflections split into two classes. In the pentagon the rotations reach all five axes, and there is one class.

In symmetric groups, conjugation is literally relabelling:

Lemma 2.8 (Relabelling). For $\sigma \in S_n$ and any cycle,

$$\sigma\,(a_1\,a_2\,\ldots\,a_k)\,\sigma^{-1} = (\sigma(a_1)\,\sigma(a_2)\,\ldots\,\sigma(a_k)).$$

Proof. Let $\tau = (a_1\,a_2\,\ldots\,a_k)$. Both sides are permutations, so it is enough to check what each side does to every point.

First take a point $\sigma(a_i)$ with $1 \le i \le k$. The left-hand side sends it to

$$(\sigma\tau\sigma^{-1})(\sigma(a_i)) = \sigma(\tau(a_i)) = \sigma(a_{i+1}),$$

the index $i+1$ read modulo k, so that a_1 follows a_k. But the cycle on the right, $(\sigma(a_1)\,\ldots\,\sigma(a_k))$, does exactly the same to $\sigma(a_i)$: it sends each of its entries $\sigma(a_i)$ to $\sigma(a_{i+1})$.

Next, take a point b not of the form $\sigma(a_i)$. Then $\sigma^{-1}(b) \notin \{a_1, \ldots, a_k\}$, so τ fixes that point, and the left-hand side fixes b as well:

$$(\sigma\tau\sigma^{-1})(b) = \sigma\big(\tau(\sigma^{-1}(b))\big) = \sigma(\sigma^{-1}(b)) = b.$$

The cycle on the right fixes b too, since b is not one of the entries of that cycle.

The two sides act identically on every point, so they are equal. ∎

Proposition 2.9 (Conjugacy in S_n). Two permutations are conjugate in S_n if and only if they have the same *cycle type*, i.e. the same multiset of cycle lengths in their disjoint cycle decompositions.

Proof. Writing π as a product of disjoint cycles and applying Lemma 2.8 cycle by cycle, $\sigma\pi\sigma^{-1}$ is the product of the relabelled cycles, which are again disjoint: the cycle type is preserved. Conversely, given π, π' of the same type, write their disjoint cycles beneath one another, matched by length (including fixed points as 1-cycles); the map σ sending each symbol of π to the symbol below it is a permutation, and by the lemma $\sigma\pi\sigma^{-1} = \pi'$. ∎

Example 2.10 (Conjugacy classes of S_3). The cycle types in S_3 are: identity; one transposition; one 3-cycle. Hence the classes are

$$S_3 \;=\; \{e\} \;\cup\; \{(1\,2), (1\,3), (2\,3)\} \;\cup\; \{(1\,2\,3), (1\,3\,2)\},$$

of sizes $1 + 3 + 2 = 6$. Under the isomorphism $D_3 \cong S_3$ of Lecture 1 these match the classes $\{e\}$, {three reflections}, $\{r, r^2\}$ — as they must: an isomorphism carries classes to classes. ◁

How large can a conjugacy class be at most? The answer is another application of the definition of cosets from Lecture 1.

Definition 2.11 (Centraliser and centre). The *centraliser* of $g \in G$ is

$$C_G(g) = \{x \in G : xg = gx\}.$$

It is a subgroup (closure and inverses are immediate). The *centre* of G is

$$Z(G) = \{x \in G : xg = gx \text{ for all } g \in G\},$$

that is, the set of elements commuting with everything. The two definitions are one definition seen from two sides: commuting with every g is exactly belonging to every

centraliser, so

$$Z(G) = \bigcap_{g \in G} C_G(g).$$

In particular $Z(G)$ is a subgroup. The group G is abelian precisely when $Z(G) = G$, and $[g] = \{g\}$ precisely when $g \in Z(G)$.

Proposition 2.12 (Class sizes). For g in a finite group G,

$$|[g]| = [G : C_G(g)].$$

In particular every class size divides $|G|$, and $|G| = \sum_i |[g_i]|$, summed over the distinct classes (the *class equation*).

Proof. The class $[g]$ consists of the elements xgx^{-1}, where x runs over the whole group G. There are $|G|$ choices of x, but different x can give the same conjugate. For example, both $x = e$ and $x = g$ give g. So we must count how many *different* elements xgx^{-1} there are.

First we find out when two elements $x, y \in G$ give the same conjugate:

$$\begin{aligned} xgx^{-1} = ygy^{-1} &\iff (y^{-1}x)\, g\, (x^{-1}y) = g \iff (y^{-1}x)\, g\, (y^{-1}x)^{-1} = g \\ &\iff y^{-1}x \in C_G(g) \iff xC_G(g) = yC_G(g). \end{aligned}$$

The first equivalence comes from multiplying both sides by y^{-1} on the left and by y on the right. The second only rewrites $x^{-1}y = (y^{-1}x)^{-1}$. The third is the definition of the centraliser, because $hgh^{-1} = g$ says the same as $hg = gh$. The fourth is the condition for equal cosets from Lecture 1: for a subgroup H, $aH = bH$ holds if and only if $a^{-1}b \in H$.

So x and y give the same conjugate if and only if they lie in the same left coset of the subgroup $C_G(g)$. In other words, the cosets sort the elements of G by the conjugate they produce. For example, take $g = s$ in D_3. Then $C_{D_3}(s) = \{e, s\}$, and the three cosets of this subgroup, $\{e, s\}$, $\{r, rs\}$ and $\{r^2, r^2s\}$, produce the three conjugates s, r^2s and rs. Hence $|[s]| = 6/2 = 3$.

Now we can count. Assign to each coset the conjugate that its elements produce:

$$xC_G(g) \longmapsto xgx^{-1}.$$

This assignment is *well defined*, because all elements of one coset give the same conjugate (the equivalences read from right to left). The assignment is *injective*, because different cosets give different conjugates (the equivalences read from left to right). The assignment is *surjective*, because by the definition of $[g]$ every element of the class has the form xgx^{-1}. So it is a bijection between the set of cosets and the class $[g]$, and $|[g]|$ equals the number of cosets $[G : C_G(g)]$. By Lagrange's theorem this number divides $|G|$.

That leaves the class equation. Conjugacy is an equivalence relation, so distinct classes do not overlap and together they cover the whole group G. Adding up their sizes gives $|G| = \sum_i |[g_i]|$. ∎

The class equations $10 = 1 + 2 + 2 + 5$ of D_5 and $6 = 1 + 3 + 2$ of S_3, together with $Z(D_5) = \{e\}$ and $Z(S_3) = \{e\}$, illustrate both assertions. For D_4, the reading's remark gives $8 = 1 + 1 + 2 + 2 + 2$. Its centre is made of the two singleton classes, $Z(D_4) = \{e, r^2\}$.

2.3 Normal subgroups and quotient groups

Which subgroups can be "forgotten", i.e. arise as kernels of homomorphisms? Kernels have a special stability property: if $k \in \ker \varphi$, then

$$\varphi(gkg^{-1}) = \varphi(g)\, e_H\, \varphi(g)^{-1} = e_H,$$

so $g(\ker\varphi)g^{-1} \subseteq \ker\varphi$ for every g. Subgroups with this property get a name.

Definition 2.13 (Normal subgroup)**.** A subgroup $N \leq G$ is *normal*, written $N \trianglelefteq G$, if $gNg^{-1} = N$ for all $g \in G$, where $gNg^{-1} = \{gng^{-1} : n \in N\}$.

Proposition 2.14 (Equivalent formulations)**.** For a subgroup $N \leq G$, the following are equivalent:

(i) $gNg^{-1} \subseteq N$ for all $g \in G$;
(ii) $gNg^{-1} = N$ for all $g \in G$;
(iii) $gN = Ng$ for all $g \in G$ (left and right cosets coincide);
(iv) N is a union of conjugacy classes of G.

Proof. (i)⇒(ii): applying (i) to g^{-1} gives $g^{-1}Ng \subseteq N$, i.e. $N \subseteq gNg^{-1}$. (ii)⇒(iii): multiply $gNg^{-1} = N$ on the right by g. (iii)⇒(i): $gn = n'g$ for some $n' \in N$, so $gng^{-1} = n' \in N$. (ii)⇔(iv): N is stable under all conjugations iff with each element it contains that element's entire class. ∎

[self-study]

Example 2.15.

(i) In an abelian group every subgroup is normal ($gng^{-1} = n$). The centre $Z(G)$ is normal in any G for the same reason.
(ii) Kernels of homomorphisms are normal, as computed above; thus $A_n \trianglelefteq S_n$ (kernel of sgn), $\mathrm{SL}(n,\mathbb{F}) \trianglelefteq \mathrm{GL}(n,\mathbb{F})$ (kernel of det), and $\langle r\rangle \trianglelefteq D_n$ (kernel of the orientation map ν).
(iii) Any subgroup of index 2 is normal: if $H \leq G$ with $[G:H] = 2$, then for $g \notin H$ both the left and the right coset of g must be the complement $G \setminus H$, so $gH = Hg$; for $g \in H$ this is trivial. This re-proves $\langle r\rangle \trianglelefteq D_n$ and $A_n \trianglelefteq S_n$ without mentioning a homomorphism.
(iv) Reflections in D_3 generate *non-normal* subgroups: $r\,s\,r^{-1} = r^2 s \notin \langle s\rangle = \{e, s\}$. Here $\{e, s\}$ is *not* a union of classes — the class of s is all three reflections (Example 2.10 and Figure 2.3). ◁

Remark 2.16 (When is a centraliser normal)**.** Conjugation is a change of reference frame, so it carries a centraliser to a centraliser:

$$h\, C_G(g)\, h^{-1} = C_G\big(hgh^{-1}\big),$$

because x commutes with g exactly when hxh^{-1} commutes with hgh^{-1}. Comparing with Definition 2.13 gives the criterion: $C_G(g)$ is normal if and only if every element of the class $[g]$ has the *same* centraliser.

Two cases where this holds automatically: if $g \in Z(G)$ then $C_G(g) = G$; and if $|[g]| = 2$ then Proposition 2.12 makes the index $[G : C_G(g)]$ equal to 2, and such a subgroup is normal by part (iii) of the reading's example.

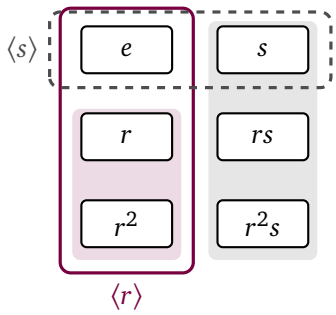


Figure 2.3: The two partitions of D_3 in one picture. The rows are the cosets of $\langle s\rangle$, the columns the cosets of $\langle r\rangle$, and the shaded regions are the conjugacy classes $\{e\}$, $\{r, r^2\}$ and $\{s, rs, r^2s\}$. The subgroup $\langle r\rangle$ (solid outline) is made of whole classes, so it is normal. The subgroup $\langle s\rangle$ (dashed outline) cuts across the class of reflections, so it is not.

Both extremes of that example are of exactly this kind. For $n \geq 3$ no reflection commutes with r, so $C_{D_n}(r) = \langle r\rangle$ — the same normal subgroup as in parts (ii) and (iii). In D_3, by contrast, $C_{D_3}(s) = \{e, s\}$, while the class $[s]$ contains all three reflections and their centralisers differ: $C_{D_3}(rs) = \{e, rs\}$. So $\{e, s\}$ is not normal — which is part (iv), seen from the other side.

Normality is exactly what is needed to multiply cosets.

Theorem 2.17 (Quotient group). Let $N \trianglelefteq G$, and let G/N denote the set of cosets of N in G (left and right cosets agreeing). Then

$$(gN)(hN) := (gh)N$$

is well defined — independent of the representatives g, h chosen — and makes G/N a group, the *quotient group*, of order $[G : N]$. The *natural projection*

$$\pi : G \to G/N, \quad \pi(g) = gN,$$

is a surjective homomorphism with $\ker \pi = N$. Conversely, if the recipe above is well defined for a subgroup N, then N is normal: the normal subgroups of G are precisely the kernels of homomorphisms out of G.

Proof. First we show that the product of cosets is well defined. Let $g' = gn$, $h' = hm$, where $n, m \in N$, be other representatives. Then

$$g'h' = gnhm = g(hh^{-1})nhm = gh(h^{-1}nh)m \in (gh)N,$$

since $h^{-1}nh \in N$ by normality; so $(g'h')N = (gh)N$. The group axioms for G/N are inherited at once from G: associativity from associativity, identity $eN = N$, inverse $(gN)^{-1} = g^{-1}N$. The number of elements is the number of cosets, $[G : N]$.

The projection satisfies $\pi(gh) = (gh)N = (gN)(hN) = \pi(g)\pi(h)$ and is onto, and $\pi(g) = N$ iff $g \in N$.

For the converse, take any $n \in N$ and $g \in G$, and compute the product of the cosets gN and $g^{-1}N$ with two choices of representative: $(g)(g^{-1}) = e$ gives the coset N, while

$(gn)(g^{-1}) = gng^{-1}$ must then also come from N. Hence $gNg^{-1} \subseteq N$ for all g, which is normality by Proposition 2.14. ∎

[self-study]

Example 2.18.

(i) $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$: the cosets of $n\mathbb{Z}$ in $\mathbb{Z}$ are the residue classes modulo n, and coset addition is addition of residues.

(ii) $D_n/\langle r\rangle$ has two elements: the coset of rotations and the coset of reflections. Its multiplication table says:

$$\text{rotation} \times \text{rotation} = \text{rotation}, \quad \text{rotation} \times \text{reflection} = \text{reflection},$$
$$\text{reflection} \times \text{reflection} = \text{rotation}.$$

That is the group C_2, and it is precisely the arithmetic of orientation. The quotient has *coarse-grained the structure of* D_n: it retains only the information the orientation homomorphism ν sees.

(iii) Likewise, for $n \geq 2$, $S_n/A_n \cong C_2$ is the arithmetic of parity:

$$\text{even} \times \text{even} = \text{even}, \quad \text{odd} \times \text{odd} = \text{even},$$
$$\text{even} \times \text{odd} = \text{odd}, \quad \text{odd} \times \text{even} = \text{odd}.$$

◁

2.4 The first isomorphism theorem

The pattern of the last example — quotient by the kernel reproduces the image — is completely general, and is perhaps the most used theorem of elementary group theory.

Theorem 2.19 (First isomorphism theorem)**.** Let $\varphi\colon G \to H$ be a homomorphism with kernel $K = \ker\varphi$. Then

$$\bar{\varphi}\colon G/K \longrightarrow \operatorname{im}\varphi, \qquad \bar{\varphi}(gK) = \varphi(g),$$

is an isomorphism: $G/\ker\varphi \cong \operatorname{im}\varphi$.

[self-study]

Proof. The formula is well defined and injective for the same reason:

$$gK = g'K \iff g^{-1}g' \in K \iff \varphi(g^{-1}g') = e_H \iff \varphi(g) = \varphi(g').$$

It is a homomorphism, $\bar{\varphi}\big((gK)(hK)\big) = \bar{\varphi}(ghK) = \varphi(gh) = \varphi(g)\varphi(h)$, and it is surjective onto $\operatorname{im}\varphi$ by construction. ∎

Example 2.20.

(i) The orientation map $\nu\colon D_n \to \{\pm 1\}$ is surjective with kernel $\langle r\rangle$, so $D_n/\langle r\rangle \cong C_2$ — now a one-line consequence rather than a table inspection. Likewise $S_n/A_n \cong C_2$ from sgn, for $n \geq 2$.

(ii) $\det\colon \mathrm{O}(n) \to \{\pm 1\}$ is surjective with kernel $\mathrm{SO}(n)$, so $\mathrm{O}(n)/\mathrm{SO}(n) \cong C_2$: orthogonal transformations split into two orientation components.

(iii) $\varphi(x) = e^{2\pi i x}$ maps $(\mathbb{R}, +)$ onto $\mathrm{U}(1)$ with kernel $\mathbb{Z}$, so

$$\mathbb{R}/\mathbb{Z} \;\cong\; \mathrm{U}(1)\,:$$

winding the real line onto the circle (Figure 2.4). Quotients of *continuous* groups will be everyday objects in Part II. ◁

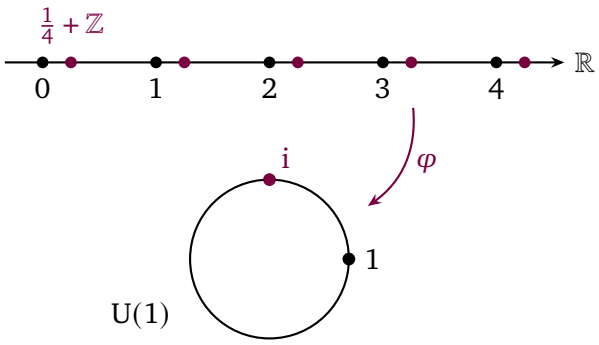


Figure 2.4: Winding the real line onto the circle, $\varphi(x) = e^{2\pi i x}$. All integers (black) go to 1, all points of $\frac{1}{4}+\mathbb{Z}$ (coloured) go to i. Each coset $x+\mathbb{Z}$ becomes a single point of the circle, so $\mathbb{R}/\mathbb{Z} \cong \mathrm{U}(1)$.

[tutorial] **Example 2.21** ($A_4/V \cong C_3$)**.** Take the rotation group A_4 of the tetrahedron, of order 12. It has four conjugacy classes:

(i) the identity;

(ii) the three half-turns $(1\,2)(3\,4)$, $(1\,3)(2\,4)$, $(1\,4)(2\,3)$ about the axes through opposite edge midpoints (Figure 2.5);

(iii) the four 3-cycles $(1\,2\,3)$, $(1\,3\,4)$, $(1\,4\,2)$, $(2\,4\,3)$ — rotations by 120° about the vertex axes;

(iv) and the four inverse 3-cycles $(1\,3\,2)$, $(1\,4\,3)$, $(1\,2\,4)$, $(2\,3\,4)$ — the same rotations the other way.

The union of the first two classes,

$$V = \{e,\ (1\,2)(3\,4),\ (1\,3)(2\,4),\ (1\,4)(2\,3)\}$$

is a subgroup, the Klein four-group. Since this subgroup is a union of classes, Proposition 2.14 makes it normal: $V \trianglelefteq A_4$. The quotient has $[A_4 : V] = 12/4 = 3$ elements, and the only group of order 3 is cyclic, so

$$A_4/V \ \cong\ C_3.$$

Each of the two nontrivial cosets of V is *exactly* one conjugacy class of 3-cycles, while V itself glues the remaining two together. The first isomorphism theorem says that the projection $\pi\colon A_4 \to A_4/V$ collapses each such slice to a single element of the quotient. ◁

2.5 Direct products *[self-study]*

We have learnt to take groups apart. Now we learn to put them together.

Definition 2.22 (Direct product)**.** The *direct product* of groups G and H is the set

$$G \times H = \{(g,h) : g \in G,\ h \in H\}$$

with componentwise multiplication $(g,h)(g',h') = (gg',hh')$. It is a group of order $|G|\,|H|$ (when both are finite).

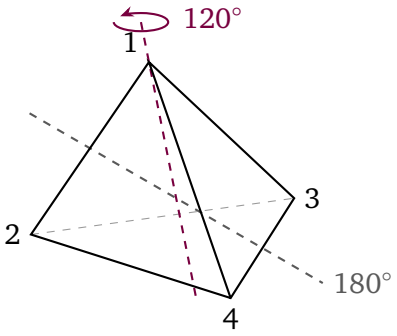


Figure 2.5: The regular tetrahedron with numbered vertices. The rotation by 120° about the axis through vertex 1 and the centre of the opposite face permutes the other three vertices cyclically: a 3-cycle. The half-turn about the axis through the midpoints of edges 12 and 34 swaps $1 \leftrightarrow 2$ and $3 \leftrightarrow 4$: it is $(1\,2)(3\,4)$.

Proposition 2.23. In $G \times H$, if g and h have finite order, then the order of (g, h) is the least common multiple of $\operatorname{ord}(g)$ and $\operatorname{ord}(h)$. If either component has infinite order, then (g, h) has infinite order. Consequently $C_m \times C_n \cong C_{mn}$ if and only if $\gcd(m, n) = 1$.

Proof. When both component orders are finite, $(g, h)^k = (g^k, h^k)$ equals (e, e) iff $\operatorname{ord}(g) \mid k$ and $\operatorname{ord}(h) \mid k$ (Lecture 1's order lemma, which needs only finite element orders), iff $\operatorname{lcm}(\operatorname{ord} g, \operatorname{ord} h) \mid k$. If either component has infinite order, no positive power of the pair can be the identity, since that would make the corresponding component have finite order. If $\gcd(m, n) = 1$ and g, h generate C_m, C_n, then (g, h) has order $\operatorname{lcm}(m, n) = mn = |C_m \times C_n|$, so the product is cyclic. If $d = \gcd(m, n) > 1$, every element's order divides $\operatorname{lcm}(m, n) = mn/d < mn$. So no element has order mn, whereas a generator of a cyclic group with mn elements would have order exactly mn. Hence the product $C_m \times C_n$ is not cyclic. ∎

Thus $C_6 \cong C_2 \times C_3$, but $C_2 \times C_2 \ncong C_4$ — the latter is the Klein four-group.

It is often useful to recognise a direct product *inside* a given group.

Proposition 2.24 (Internal direct product)**.** Let $N, M \trianglelefteq G$ with $N \cap M = \{e\}$ and

$$NM := \{nm : n \in N,\ m \in M\} = G.$$

Then every element of N commutes with every element of M, and

$$N \times M \longrightarrow G, \qquad (n, m) \longmapsto nm$$

is an isomorphism.

Proof. For $n \in N$, $m \in M$, consider the *commutator* $c = nmn^{-1}m^{-1}$. Reading it as $(nmn^{-1})m^{-1}$, normality of M puts it in M; reading it as $n(mn^{-1}m^{-1})$, normality of N puts it in N. So $c \in N \cap M = \{e\}$, i.e. $nm = mn$. The map is then a homomorphism: $(nm)(n'm') = nn'mm'$ by the commuting just proved. It is surjective since $NM = G$, and injective since $nm = e$ forces $n = m^{-1} \in N \cap M = \{e\}$. ∎

Example 2.25 (Parity: $\mathrm{O}(3) \cong \mathrm{SO}(3) \times C_2$)**.** In three dimensions let $-\mathbb{1}$ be the *spatial inversion* (parity operation) $\vec{r} \mapsto -\vec{r}$. It commutes with every matrix, so $M = \{\mathbb{1}, -\mathbb{1}\} \cong C_2$ is a normal

subgroup of $\mathrm{O}(3)$. The subgroup $N = \mathrm{SO}(3)$ is normal too, because it is the kernel of the determinant: $\mathrm{SO}(3) = \{A \in \mathrm{O}(3) : \det A = 1\}$. Since $\det(-\mathbb{1}) = (-1)^3 = -1$, inversion is *not* a rotation: $N \cap M = \{\mathbb{1}\}$. Finally every $A \in \mathrm{O}(3)$ with $\det A = -1$ factors as $A = (-\mathbb{1})(-A)$ with $-A \in \mathrm{SO}(3)$, so $NM = \mathrm{O}(3)$. By Proposition 2.24,

$$\mathrm{O}(3) \;\cong\; \mathrm{SO}(3) \times C_2.$$

Every orthogonal transformation of space is uniquely a rotation, possibly followed by inversion — the group-theoretic statement that *parity* is an independent, commuting label on top of rotational symmetry. (This will let us treat the parity of quantum states as a separate quantum number in Part II.) The argument visibly uses that the dimension is odd: in the plane, $-\mathbb{1}$ is the rotation through π, it *does* lie in $\mathrm{SO}(2)$, and no such splitting of $\mathrm{O}(2)$ exists. ◁

Direct products also describe physical systems with independent symmetries — two non-interacting subsystems, or geometric and internal symmetries acting on different attributes of the same object.

Semidirect products

The internal direct product asked both factors to be normal. Dropping that demand from one of them gives a strictly larger and far more common construction — the one behind every symmetry group of a space with a distinguished origin.

Definition 2.26 (Internal semidirect product)**.** Let $N \trianglelefteq G$ and $H \leq G$ with $N \cap H = \{e\}$ and $NH = G$. Then G is the *(internal) semidirect product* of N by H, written

$$G = N \rtimes H.$$

Each $g \in G$ factors uniquely as $g = nh$ ($n \in N$, $h \in H$): existence is $NH = G$, and if $nh = n'h'$ then $n'^{-1}n = h'h^{-1} \in N \cap H = \{e\}$. Moreover conjugation by H stabilises N: $hnh^{-1} \in N$ for all $h \in H,\ n \in N$.

Only N is required normal; H need not be. When H *is* also normal the conjugation action is trivial and the construction collapses back to the direct product of Proposition 2.24.

An *automorphism* of N is an isomorphism from N to itself. The automorphisms form a group under composition, denoted $\mathrm{Aut}(N)$.

An *action of H on N by automorphisms* means exactly a homomorphism $\varphi\colon H \to \mathrm{Aut}(N)$; writing φ_h for the image of h, the two laws packed into that word are

$$\varphi_e = \mathrm{id}_N, \qquad \varphi_{hh'} = \varphi_h \circ \varphi_{h'}.$$

In words: each h twists N by an automorphism, the identity twists nothing, and twisting by a product is twisting twice. This twist is what the next construction inserts into an otherwise ordinary product of pairs.

Proposition 2.27 (External model)**.** Given groups N, H and a homomorphism $\varphi\colon H \to \mathrm{Aut}(N)$, written $h \mapsto \varphi_h$, the set $N \times H$ with

$$(n, h)(n', h') = \big(n\,\varphi_h(n'),\ hh'\big)$$

is a group $N \rtimes_\varphi H$, with identity (e,e) and inverse $(n,h)^{-1} = \big(\varphi_{h^{-1}}(n^{-1}),\, h^{-1}\big)$. Identifying N with $N \times \{e\}$ and H with $\{e\} \times H$ realises Definition 2.26, the action being $\varphi_h(n) = hnh^{-1}$.

Proof. Associativity is the single computation

$$\begin{aligned}\big((n,h)(n',h')\big)(n'',h'') &= \big(n\,\varphi_h(n')\,\varphi_{hh'}(n''),\; hh'h''\big)\\ &= (n,h)\big((n',h')(n'',h'')\big),\end{aligned}$$

where $\varphi_{hh'} = \varphi_h \circ \varphi_{h'}$ (homomorphism) and each $\varphi_h \in \mathrm{Aut}(N)$ distributes over products. The identity and inverse are immediate. ∎

Example 2.28 (The dihedral group)**.** $D_n = \langle r, s \mid r^n = s^2 = e,\ srs^{-1} = r^{-1}\rangle$ has the rotation subgroup $\langle r\rangle \cong C_n$ normal of index two and the reflection $\langle s\rangle \cong C_2$ meeting it trivially, so $D_n \cong C_n \rtimes C_2$ with C_2 acting on C_n by inversion, $\varphi_s(r) = r^{-1}$.

It is worth running the ordered-pair rule once. Writing elements as (r^a, s^ε), the product of Proposition 2.27 gives

$$(r^a,s^\varepsilon)(r^b,s^\delta) = \big(r^a\,\varphi_{s^\varepsilon}(r^b),\; s^{\varepsilon+\delta}\big) = \big(r^{a+(-1)^\varepsilon b},\; s^{\varepsilon+\delta}\big),$$

which is exactly the multiplication rule for D_n from Lecture 1. The smallest instance shows the twist on its own: $(e,s)(r,e) = (\varphi_s(r), s) = (r^{-1}, s)$ — passing r across s inverts it. The action is nontrivial for $n \geq 3$: dihedral groups are genuinely semidirect, not direct. ◁

The construction carries over unchanged from finite groups to the continuous groups of Part II. The symmetry group of the plane is the Euclidean group $\mathrm{E}(2) = \mathbb{R}^2 \rtimes \mathrm{O}(2)$ (the translations are normal, acted on by rotations and reflections), and the symmetry group of relativistic spacetime is the Poincaré group $\mathbb{R}^4 \rtimes \mathrm{SO}^+(1,3)$ of Lecture 17 — translations normal, acted on by Lorentz transformations. In both cases every symmetry is written uniquely as a transformation that fixes the origin (a rotation, a reflection or a Lorentz transformation) followed by a translation. In the plane this is the map $\vec{r} \mapsto A\vec{r} + \vec{b}$: first $A \in \mathrm{O}(2)$, then the translation by $\vec{b}$. This is exactly the unique decomposition $g = nh$ of Definition 2.26.

2.6 Simple groups and composition series *[self-study]*

Quotients suggest a notion of *atom*: a group that cannot be coarse-grained at all.

Definition 2.29 (Simple group)**.** A group G with $|G| > 1$ is *simple* if its only normal subgroups are $\{e\}$ and G. Equivalently, every nontrivial homomorphism out of G is injective.

Proposition 2.30. An abelian group is simple if and only if it is cyclic of prime order.

Proof. By Lagrange's theorem, C_p (with p prime) has no subgroups at all besides $\{e\}$ and itself. Conversely, in an abelian group every subgroup is normal, so a simple abelian G may have no proper nontrivial subgroup whatsoever. Pick $g \neq e$: then $\langle g\rangle = G$, and g cannot have infinite order (else $\langle g^2\rangle$ would be proper and nontrivial), so G is cyclic of finite order n; and if $n = ab$ with $1 < a < n$ then $\langle g^a\rangle$ is proper and nontrivial. Hence n is prime. ∎

Nonabelian simple groups exist and are plentiful: the alternating groups A_n are simple for all $n \geq 5$ (we shall not prove this). The smallest, A_5 of order 60 — the rotation group of the icosahedron — is the smallest nonabelian simple group of all. Its simplicity is, through Galois's theory, precisely the reason no formula in radicals solves the general quintic equation.

Definition 2.31 (Composition series)**.** A *composition series* of a finite group G is a chain

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_k = \{e\},$$

in which each G_{i+1} is a normal subgroup of G_i (but not necessarily a normal subgroup of G) and each *composition factor* G_i/G_{i+1} is simple.

Every finite group has a composition series. Take any chain of this form and refine it until no further proper normal subgroup can be inserted. The refinement stops because G is finite. When it stops, every factor G_i/G_{i+1} is simple. For suppose Q were a proper nontrivial normal subgroup of G_i/G_{i+1}, and let $\pi\colon G_i \to G_i/G_{i+1}$ be the projection. Then $\pi^{-1}(Q)$ is normal in G_i: for $g \in G_i$ and $x \in \pi^{-1}(Q)$, $\pi(gxg^{-1}) = \pi(g)\pi(x)\pi(g)^{-1} \in Q$ because Q is normal. And $G_{i+1} \subsetneq \pi^{-1}(Q) \subsetneq G_i$, since Q is nontrivial and proper. So the chain could be refined further, contrary to maximality.

Example 2.32 (Composition series of D_4)**.** Consider

$$D_4 \trianglerighteq \langle r \rangle \trianglerighteq \langle r^2 \rangle \trianglerighteq \{e\}.$$

Each step has index 2, hence is normal in the preceding group by Example 2.15(iii), and each factor is a group of order 2:

$$D_4/\langle r \rangle \cong C_2, \qquad \langle r \rangle/\langle r^2 \rangle \cong C_2, \qquad \langle r^2 \rangle/\{e\} \cong C_2,$$

all simple. A quite different-looking series uses the Klein subgroup $V_1 = \{e, r^2, s, r^2 s\}$ of Sheet 1:

$$D_4 \trianglerighteq V_1 \trianglerighteq \langle s \rangle \trianglerighteq \{e\},$$

again with all indices 2 ($\langle s \rangle$ is normal in the *abelian* group V_1, though not in D_4!). Both series have composition factors C_2, C_2, C_2. ◁

Example 2.33 (C_{12})**.** For $C_{12} = \langle g \rangle$, the chains

$$C_{12} \trianglerighteq \langle g^2 \rangle \trianglerighteq \langle g^4 \rangle \trianglerighteq \{e\} \quad \text{and} \quad C_{12} \trianglerighteq \langle g^3 \rangle \trianglerighteq \langle g^6 \rangle \trianglerighteq \{e\}$$

are composition series with factors C_2, C_2, C_3 and C_3, C_2, C_2 respectively: the factors arrive in a different *order*, but as a multiset they agree — the "prime factorisation" $12 = 2 \cdot 2 \cdot 3$ in group-theoretic form. ◁

This stability is general:

Theorem 2.34 (Jordan–Hölder; statement only)**.** Any two composition series of a finite group have the same length and the same composition factors with the same multiplicities, up to reordering and isomorphism.

Finite groups are thus built from simple groups as molecules from atoms, and the classification of finite simple groups mentioned in Lecture 1 is, as it were, the periodic table. But the analogy has a sting: knowing the atoms does not determine the molecule. C_4 and the Klein four-group both have composition factors C_2, C_2, yet are not isomorphic — assembling a group from given factors (the *extension problem*) has in general many inequivalent solutions, and no complete classification of all finite groups exists or is expected.

Summary.

- *Homomorphism.* A product-preserving map; its kernel is normal and its image a subgroup (Definition 2.1, Proposition 2.3).
- *Conjugacy.* Conjugacy classes partition the group; in S_n they are the cycle types (Proposition 2.9), and class sizes divide $|G|$ (Proposition 2.12).
- *Quotients.* A normal subgroup yields a quotient group (Theorem 2.17); the first isomorphism theorem gives $G/\ker\varphi \cong \operatorname{im}\varphi$ (Theorem 2.19).
- *Building groups.* The reading puts groups back together — direct products combine independent factors, and semidirect products $N \rtimes_\varphi H$ encode an action $\varphi\colon H \to \operatorname{Aut}(N)$ — and takes a finite group apart into its simple constituents.

References

Armstrong, *Groups and Symmetry* (cosets, quotient groups, direct and semidirect products); Jones, *Groups, Representations and Physics*, Ch. 2; Tung, *Group Theory in Physics*, §§2.4–2.7.